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CCS3440 - Artificial Intelligence

Assignment 02

Technical Report: Patient Readmission Prediction

Repository Link: [omega-u20/SmartCare-HospitalManagement](https://github.com/omega-u20/SmartCare-HospitalManagement)

Drive Link: <https://drive.google.com/drive/folders/1MJ8nvWKhyilassXyrSipSY0SHbIQbLJ5>

Prepared by: Group 16

Prediction Task Selected: Option B – Patient Readmission Prediction (30 days)

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1. Introduction

Hospital readmissions represent a significant clinical and operational challenge. Patients returning to the hospital shortly after discharge can indicate incomplete initial care and incur high financial costs. Predicting readmission risk in advance allows hospital management to intervene early and adjust discharge planning to potentially reduce avoidable readmissions.

2. Problem Definition & Prediction Point

This project formulates the challenge as a binary classification problem: predicting whether a patient will be readmitted to the hospital within 30 days of their original discharge (`readmitted_30_days = 1`) or not (`readmitted_30_days = 0`).

Prediction Point: To prevent trivial separation, the model's prediction point is strictly defined as applying only to *discharged inpatients*. Including records of non-admitted patients (outpatient visits) would make the task artificially easy, as non-admitted patients inherently lack inpatient discharge procedures and are evaluated on different clinical pathways.

3. Literature Review

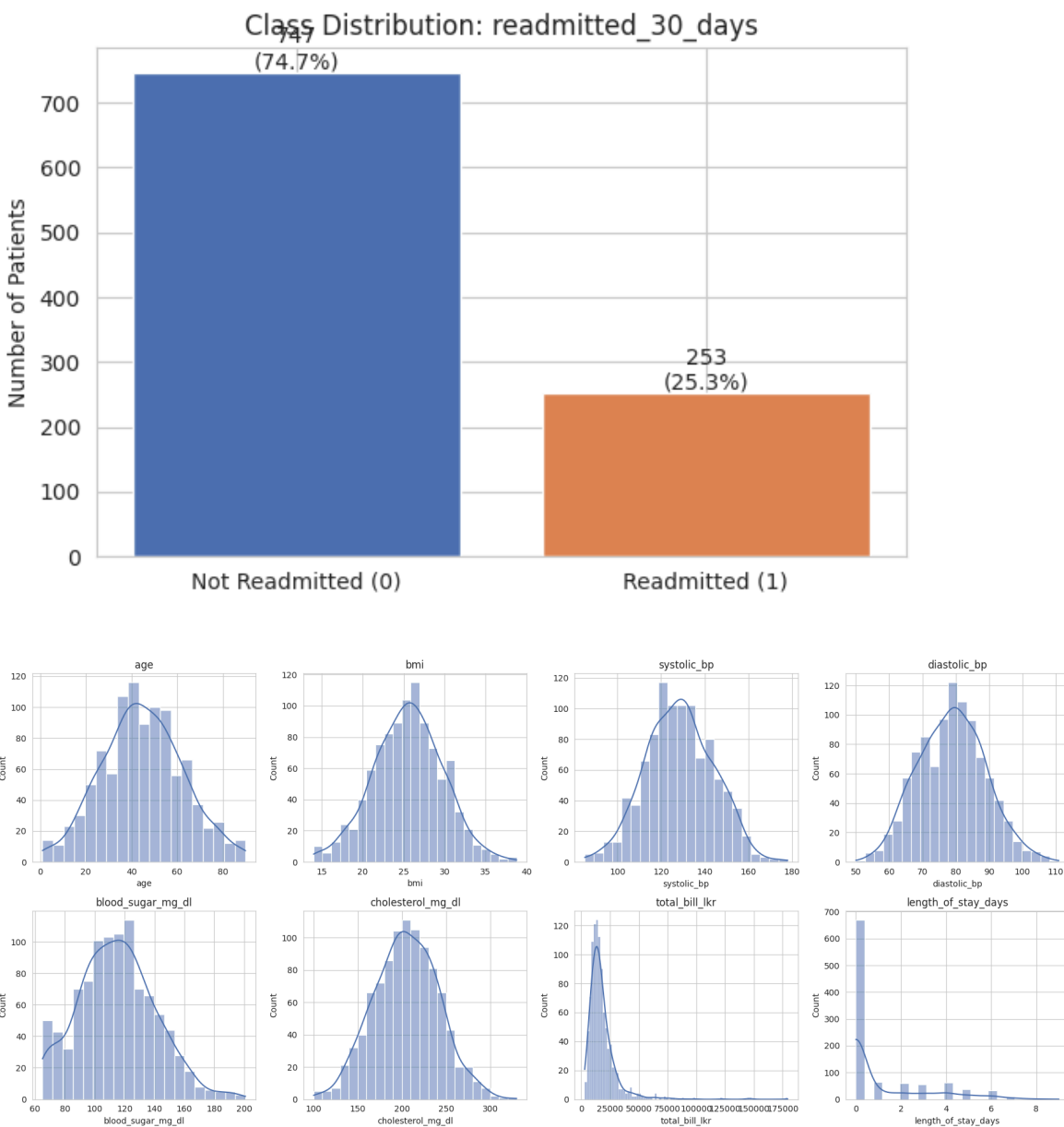
Previous studies indicate a trend towards using sophisticated deep learning and graph-based methods for predicting hospital readmissions over traditional machine learning models. While these techniques yield strong predictive outcomes, they face drawbacks such as limited external validation, class imbalance issues, and a lack of model explainability.

- **Wang et al. (2018):** Cost-Sensitive Deep Learning. Achieved an AUC of 0.70, but functions as a black-box model and relies heavily on ICU data.
- **Shao et al. (2024):** Transformer + CNN. High accuracy but restricted to a single disease cohort (Congestive Heart Failure).
- **da Silva et al. (2023):** Classical ML (CART). Good performance using classical ML, but limited exclusively to pediatric patients.
- **Lu et al. (2021):** Graph Convolutional Networks. Outperformed baselines but suffers from high computational complexity.
- **Gao et al. (2024):** Semantic Learning (CHSLM). Strong results but untested across different external hospital contexts.

4. Dataset Description & Exploratory Data Analysis

The SmartCare Hospital AI Dataset contains 1,000 patient records across 33 initial variables, covering patient demographics, appointment details, clinical measurements, and financial billing.

A class distribution analysis revealed a moderate imbalance: 74.7% of patients were not readmitted (747 records), and 25.3% were readmitted (253 records). Histograms of clinical vitals (e.g., blood pressure, BMI, cholesterol) showed relatively normal distributions, indicating genuine clinical variations rather than data errors. Financial features like total bills were right-skewed, which is a standard distribution pattern in healthcare cost data.



5. Data Preprocessing & Leakage Prevention

To ensure robust evaluation and prevent data leakage, the dataset was explicitly split into training and testing sets *before* applying any scaling, encoding, or imputation rules.

- **Train/Test Split:** An 80/20 split was applied using a random seed (random_state=42) with strict target stratification (stratify=y) to maintain the class balance.
- **Class Distributions:** The training set (800 records) contained a balance of 74.8% negative and 25.2% positive cases. The held-out test set (200 records) preserved this with 74.5% negative and 25.5% positive cases.
- **Missing Values:** The room_type column (90.6% missing rate) was structurally missing for unadmitted patients. These values were encoded as 'Not Admitted'.
- **Feature Engineering:** Features such as missed_appointment_rate, an is_hypertensive flag, and avg_charge_per_treatment were engineered.
- **Encoding & Scaling:** Categorical features were processed using One-Hot Encoding (drop_first=True). Numeric features were standardized using a StandardScaler fitted *exclusively* on the training set and then applied to transform the test set.
- **Final Feature List:** The final processed dataset contained 58 features (e.g., age, length_of_stay_days, total_bill_lkr, one-hot encoded diagnosis, blood_group, and department variants).

6. Model Development & Hyperparameter Tuning

Model selection was driven strictly by 5-fold cross-validation (CV) on the training set, optimizing for the F1-score to accommodate the class imbalance. The test set was fully reserved for the final, unbiased evaluation.

Hyperparameter Grids & Best CV Parameters:

- **Random Forest:** Grid: n_estimators [100, 200], max_depth [5, 10, None], min_samples_leaf [1, 5]. Best Parameters: max_depth=10, min_samples_leaf=1, n_estimators=100. Best CV F1: 0.883.
- **Decision Tree:** Grid: max_depth [3, 5, 7, 10, None], min_samples_leaf [1, 5, 10]. Best Parameters: max_depth=5, min_samples_leaf=10. Best CV F1: 0.883.
- **XGBoost:** Grid: n_estimators [100, 200], max_depth [3, 5, 7], learning_rate [0.05, 0.1]. Scaled positive weight: 2.96. Best Parameters: learning_rate=0.05, max_depth=3, n_estimators=200. Best CV F1: 0.884.
- **K-Nearest Neighbors:** Grid: n_neighbors [3, 5, 7, 9, 11]. Best K: 7. Best CV F1: 0.806.

7. Model Evaluation

The models were evaluated on the 200 held-out test samples. Recall and F1-score were prioritized over raw accuracy because failing to identify a readmitted patient (false negative) carries a higher clinical cost than a false positive.

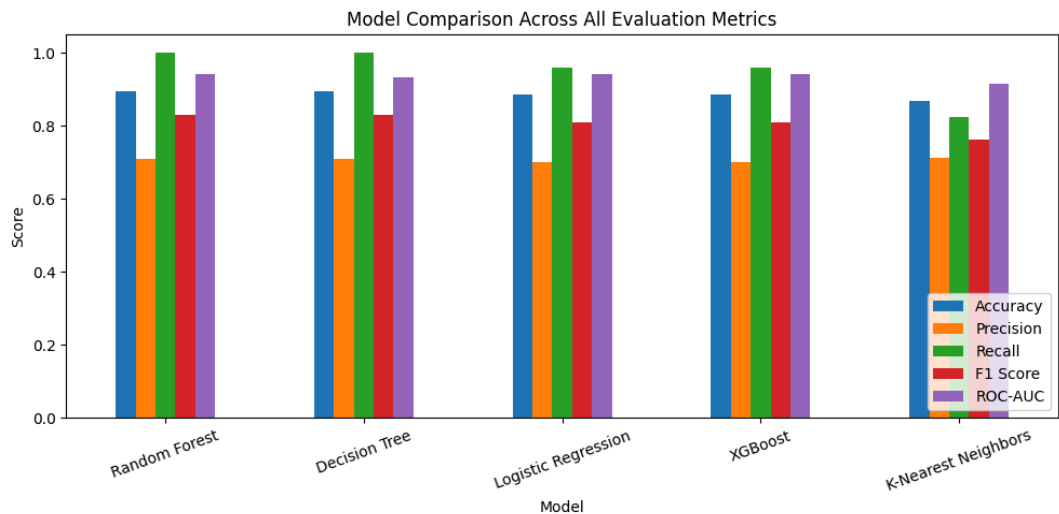
Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Random Forest	0.895	0.708	1.000	0.829	0.942
Decision Tree	0.895	0.708	1.000	0.829	0.932
Logistic Regression	0.885	0.700	0.961	0.810	0.942
XGBoost	0.885	0.700	0.961	0.810	0.942
K-Nearest Neighbors	0.870	0.712	0.824	0.764	0.915

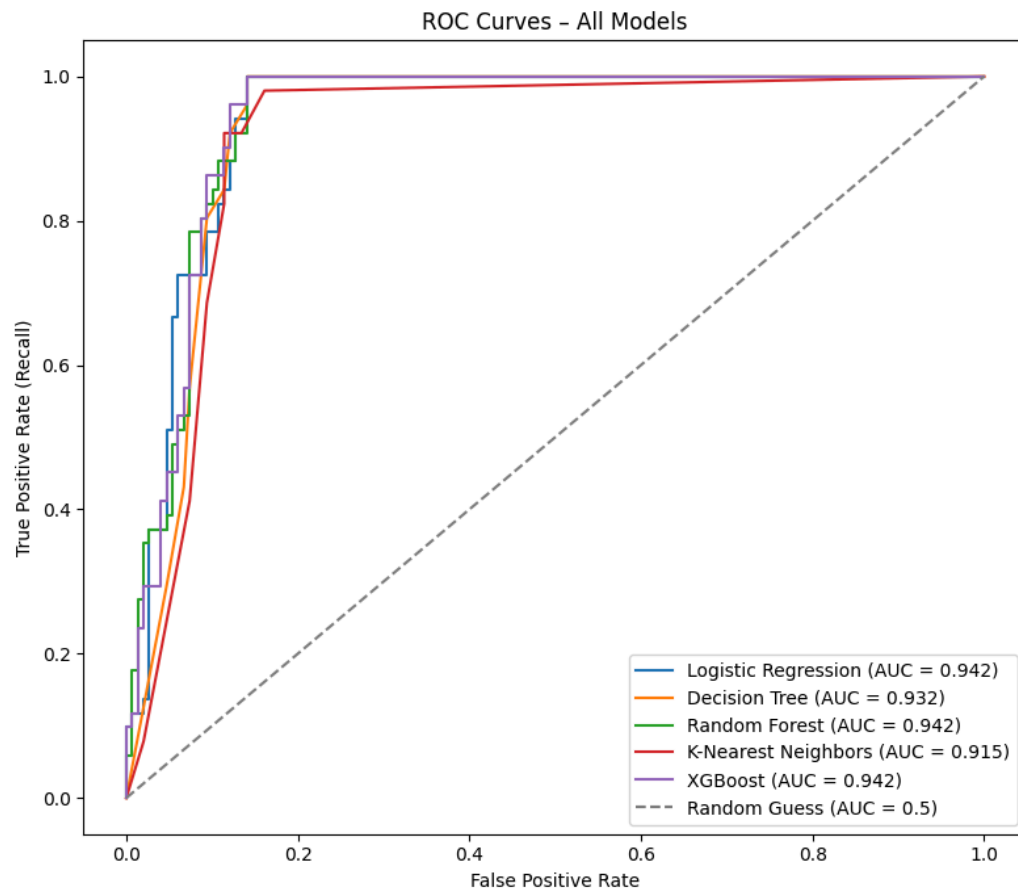
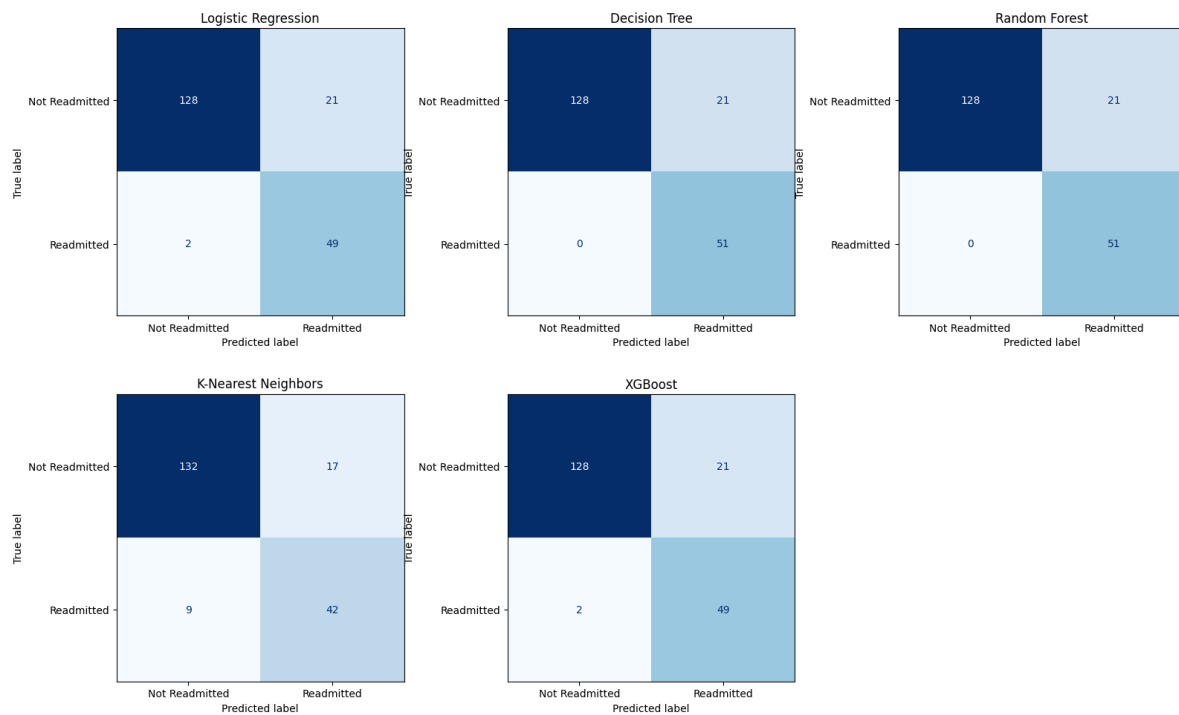
Note: All data pulled from test set unbiased evaluations.

Performance Breakdown:

The Random Forest model was selected as the final model due to its high test F1-score (0.829). While Random Forest, Logistic Regression, and XGBoost all achieved an identical ROC-AUC of 0.942 (when rounded to three decimal places), Random Forest successfully captured a recall of 1.000.

It is vital to note that while the model achieved 100% recall on the 51 positive cases within this specific test sample, this does not guarantee zero missed high-risk patients in a real-world clinical setting. Furthermore, a precision of 0.708 means the model generates false-positive alerts. For a complete clinical evaluation, future reporting should extend to metrics such as Specificity, PR-AUC, Balanced Accuracy, and MCC to fully contextualize the false-positive burden on hospital staff.

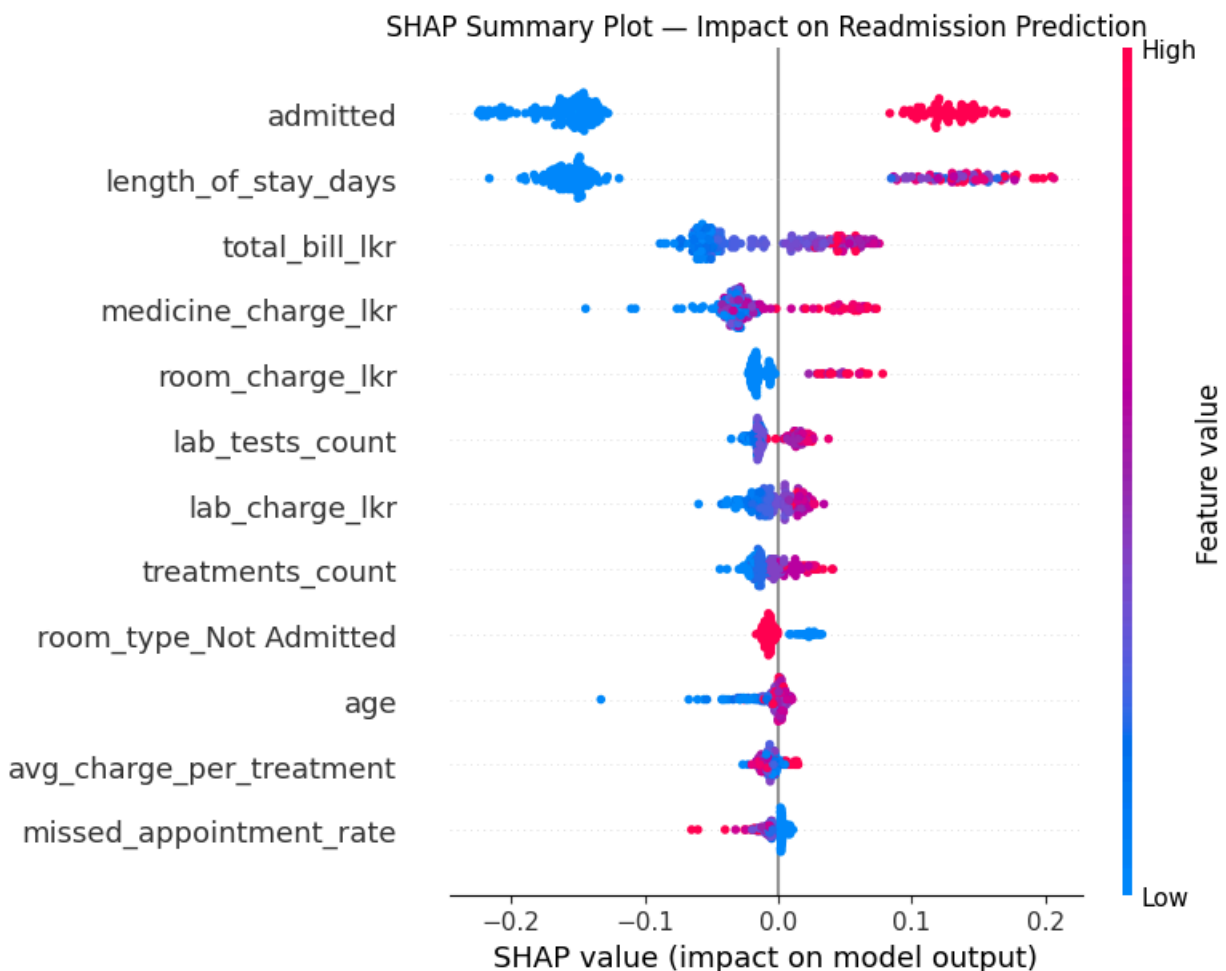


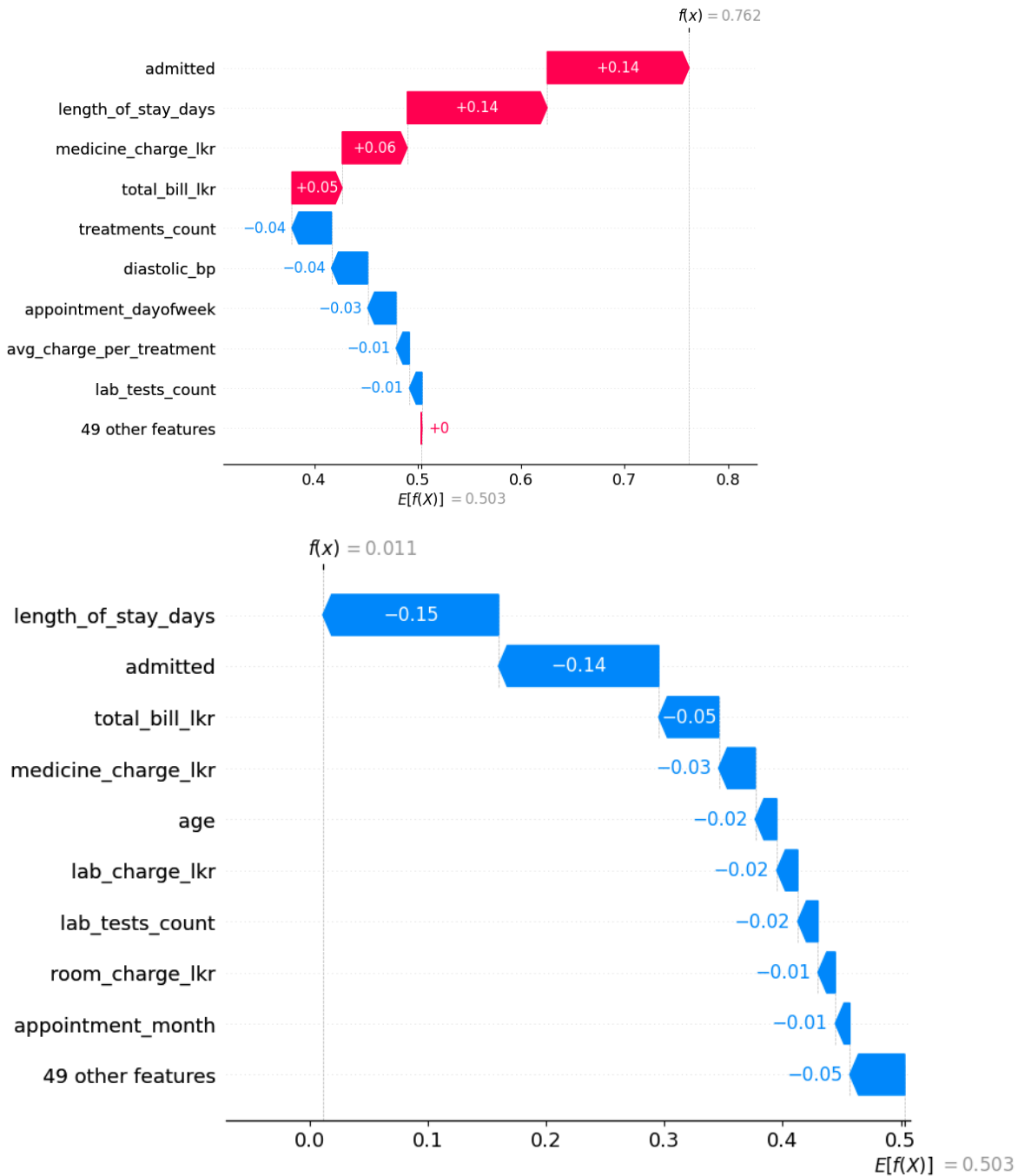


8. Explainable AI Analysis

To ensure transparency, SHAP (SHapley Additive exPlanations) values were used to explain the Random Forest model's predictions via a TreeExplainer.

- **Global Explainability:** The SHAP summary plot indicated that admitted and length_of_stay_days were the most influential predictors.
- **Target Construction Artefacts:** Because non-admitted patients were included in the current dataset, these features may act as target-construction artefacts. Patients with a length of stay of 0 days were rarely readmitted, creating a trivial mathematical separation that must be addressed in future iterations by restricting the training cohort exclusively to discharged inpatients.
- **Local Explainability:** SHAP waterfall plots were generated to show clinicians exactly how specific feature values (e.g., high total bills or specific diagnoses) contributed to an individual's specific readmission probability.





9. Prototype Design & Deployment

An AI decision-support prototype was built using Streamlit. The prototype is strictly configured to load the exact pre-fitted model artifacts (random_forest.joblib), along with the identical scaling objects (scaler.joblib), and feature column dictionaries to maintain perfect alignment with the notebook's preprocessing pipeline.

Execution & Disclaimers:

The application interface accepts numeric and categorical input arrays and validates them against the trained feature hierarchy. Predictions generated in the UI mirror the exact outputs of the notebook prototype.

Disclaimer: This application is a Version 1.0 academic prototype for decision-support testing only and is not a certified clinical or medical device.

The screenshot displays a web application titled "SmartCare Hospital: AI Decision Support" with a subtitle "30-Day Patient Readmission Prediction". The interface prompts the user to "Enter patient details below to generate a readmission risk prediction." It features a form with the following inputs:

Patient Details	
Age	Number of Treatments
45	2
Admitted During Visit (0-No, 1-Yes)	Medicine Charge (LKR)
0	5000.00
Length of Stay (Days)	Lab Tests Count
3	3
Total Bill (LKR)	Lab Charge (LKR)
15000.00	2000.00

A "Generate Prediction" button is located below the form. The "Prediction Results" section shows a green bar indicating "Low Risk of Readmission (Probability: 34.0%)" and a blue bar with the recommendation: "Recommendation: Proceed with standard discharge protocols."

10. Conclusion

The project successfully built a machine learning pipeline to predict 30-day patient readmissions for SmartCare Hospital. By rigorously preventing data leakage, comparing multiple algorithms, and integrating explainable AI, the model serves as an effective baseline tool. However, clinical plausibility does not replace rigorous statistical testing; comprehensive external validation, calibration analysis, and clinician evaluation are required before deployment in a live healthcare environment.

Key Findings and System Performance

The baseline evaluation demonstrated that ensemble tree-based architectures, specifically Random Forest, provided robust predictive capabilities for identifying high-risk readmissions prior to patient discharge. By applying strict stratification during the train-test split and standardizing feature scaling solely on the training partition, the model architecture effectively mitigated data leakage. Furthermore, the integration of SHAP framework interpretability allowed for both global feature prioritization and localized instance explanations, offering transparency regarding the key clinical and operational indicators driving risk scores.

Future Improvements and Clinical Integration

To transition this decision-support prototype into a production-ready clinical workflow, several crucial enhancements must be addressed:

- **Cohort Refinement:** Restrict the model training population exclusively to discharged inpatients to eliminate target-construction artifacts associated with non-admitted or outpatient records.
- **Model Calibration & Threshold Tuning:** Perform probability calibration (e.g., Platt scaling or isotonic regression) and optimize decision thresholds to strike an operational balance between sensitivity and false-positive alert fatigue for hospital staff.
- **Expanded Clinical Metrics:** Extend prospective evaluations to include Precision-Recall AUC (PR-AUC), Specificity, and Matthews Correlation Coefficient (MCC) alongside longitudinal external validation across partner healthcare facilities.
- **EHR System Integration:** Embed real-time inference triggers directly into Electronic Health Record (EHR) workflows to provide actionable discharge planning recommendations to attending clinicians.

11. References

- da Silva, N. C., et al. (2023). Prediction of readmissions in hospitalized children and adolescents by machine learning. *Proceedings of the 38th ACM/SIGAPP Symposium on Applied Computing*.
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